Experiment 9: Air Quality Analysis using Time Series Trends and Clustering

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Aim

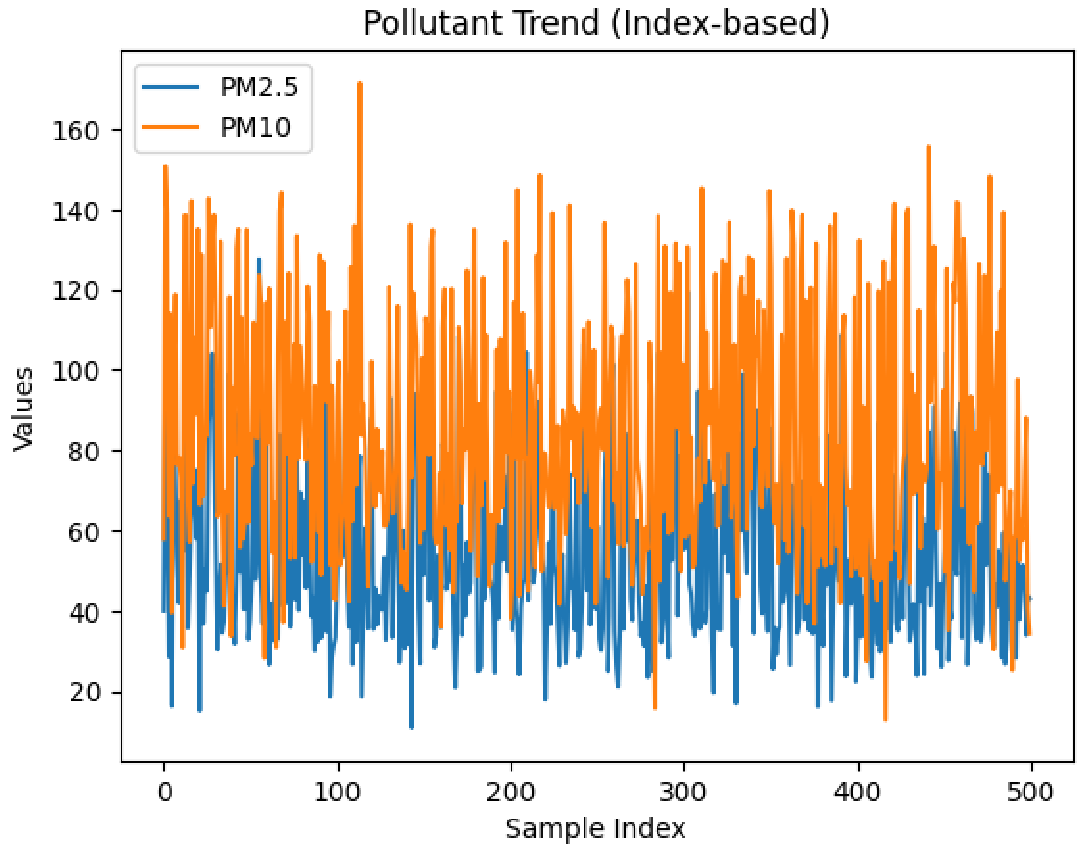
To analyze air quality data using trend analysis and moving averages, and apply K-Means clustering to group similar pollution patterns based on multiple environmental features.

Code

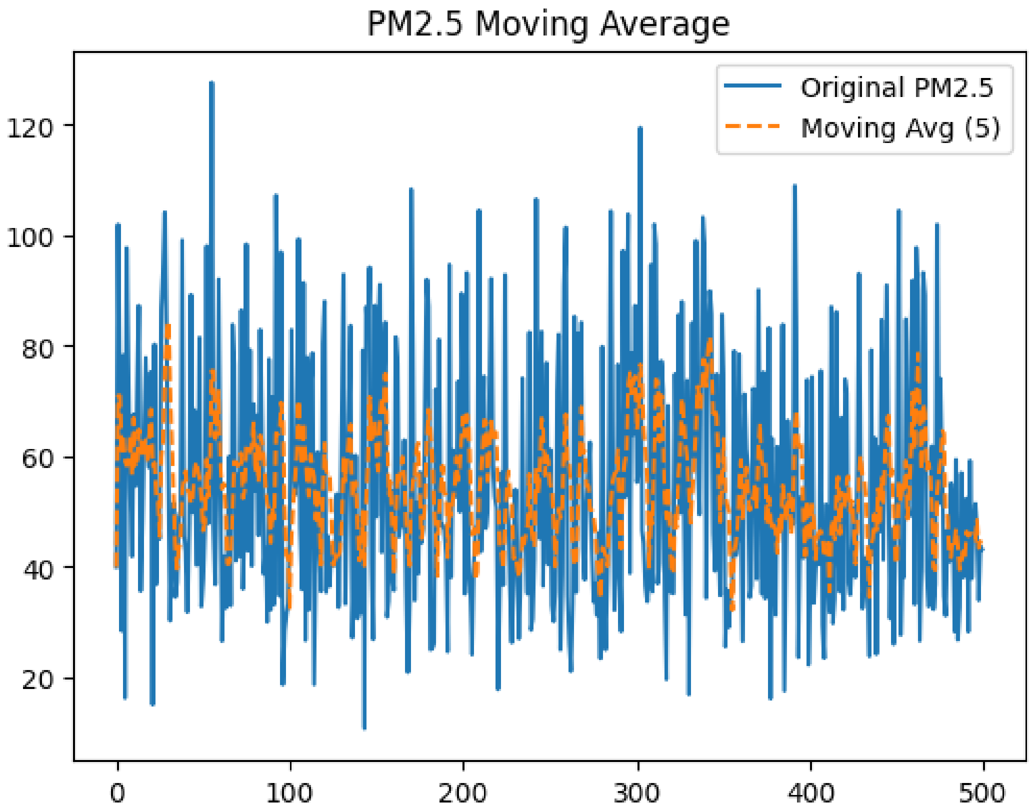
```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.cluster import KMeans


# Step 1: Load dataset

df = pd.read_csv('/content/air_quality_dataset.csv')


# Step 2: Select numerical columns

features = ['PM2.5', 'PM10', 'NOx', 'NO2', 'SO2', 'VOCs',

        'CO', 'CO2', 'CH4', 'Temperature', 'Humidity']


df_numeric = df[features]


# Step 3: Trend (Index-based)


plt.figure()

plt.plot(df_numeric['PM2.5'], label='PM2.5')

plt.plot(df_numeric['PM10'], label='PM10')

plt.legend()

plt.title("Pollutant Trend (Index-based)")

plt.xlabel("Sample Index")

plt.ylabel("Values")

plt.show()
```

```
# Step 4: Moving Average

df['PM2.5_MA'] = df['PM2.5'].rolling(window=5, min_periods=1).mean()

plt.figure()
plt.plot(df['PM2.5'], label='Original PM2.5')
plt.plot(df['PM2.5_MA'], label='Moving Avg (5)', linestyle='--')
plt.legend()
plt.title("PM2.5 Moving Average")
plt.show()

# Step 5: Clustering (MULTI-FEATURE - IMPORTANT)

kmeans = KMeans(n_clusters=3, random_state=0)
df['Cluster'] = kmeans.fit_predict(df_numeric)

# Step 6: Visualize Clusters

plt.figure()
plt.scatter(df['PM2.5'], df['PM10'], c=df['Cluster'])
plt.xlabel("PM2.5")
plt.ylabel("PM10")
plt.title("AQI Clustering (PM2.5 vs PM10)")
plt.show()

# Step 7: Cluster Analysis

print(df.groupby('Cluster')[features].mean())

print(df.head())
```

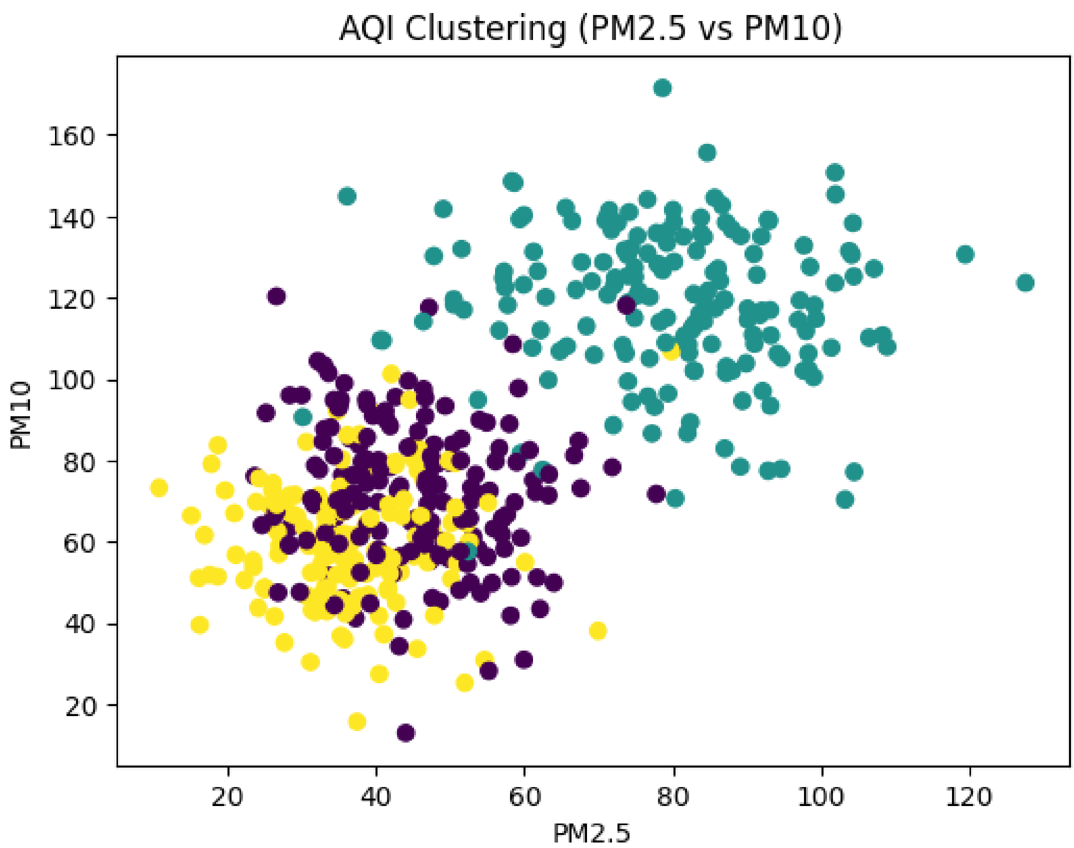
Output



Pollutant Trend (Index-based)
PM2.5
PM10
Values
Sample Index
0
100
200
300
400
500
20
40
60
80
100
120
140
160



PM2.5 Moving Average
Original PM2.5
Moving Avg (5)
0
100
200
300
400
500
20
40
60
80
100
120



Cluster	PM2.5	PM10	NOx	NO2	SO2	VOCs \
0	44.960683	71.489380	29.505894	15.430882	8.066823	99.576008
1	79.660034	118.249337	146.999039	71.291745	20.003434	151.484086
2	36.210230	60.131715	99.967161	39.938341	5.388022	79.989716

Cluster	CO	CO2	CH4	Temperature	Humidity
0	1.504787	390.818338	1.987365	28.471025	64.56232
1	2.979838	502.984149	2.528758	32.766567	48.87192
2	2.017541	420.624636	1.822129	30.024652	55.00108

	PM2.5	PM10	NOx	NO2	SO2	VOCs \
0	39.967142	57.926035	116.192213	55.230299	4.531693	75.317261
1	101.935672	150.774299	76.826826	79.051618	18.744780	145.083987
2	70.996192	138.948796	158.731020	60.466604	14.892239	145.147338
3	28.464728	63.643900	25.385343	15.333286	7.647429	130.022319
4	78.265276	113.977926	105.644340	59.202337	17.696806	181.713667

	CO	CO2	CH4	Temperature	Humidity	Wind_Direction \
0	2.789606	427.674347	1.706105	31.085120	45.454749	276
1	1.966569	529.739619	2.492663	33.711103	60.798212	134
2	2.626446	499.889443	2.431165	33.778698	54.875669	1
3	1.779360	388.283712	1.818563	31.565877	67.113319	251
4	3.240533	464.739197	2.597225	32.229835	37.236519	326

	Location_Type	Source_Label	PM2.5_MA	Cluster
0	Urban	Vehicular	39.967142	2
1	Industrial	Industrial	70.951407	1
2	Industrial	Industrial	70.966335	1
3	Rural	Biomass Burning	60.340933	0
4	Industrial	Industrial	63.925802	1

Result

The air quality dataset was analyzed to observe pollutant trends and smooth variations using moving averages. K-Means clustering was applied on multiple environmental features to group similar air quality patterns. The clusters revealed distinct pollution levels and environmental conditions, helping in better understanding and classification of air quality data.